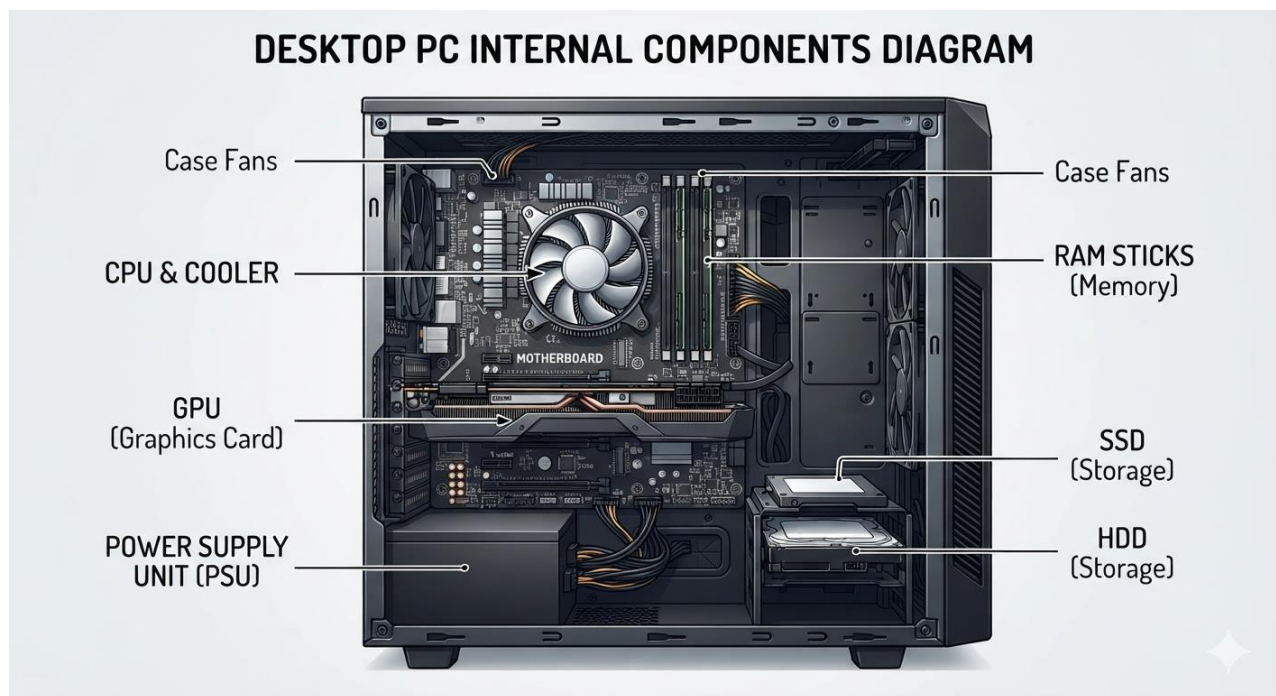


ASSIGNMENT 1.1

Q1-- Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.

ANS.



PURPOSE OF COMPONENTS: -

1. Motherboard

- The **motherboard** is the main circuit board of the computer.
- It connects all the computer parts together so they can communicate.

2. CPU (Central Processing Unit)

- The **CPU** is called the **brain of the computer**.
- It processes instructions and performs calculations to run programs and games.

3. RAM (Random Access Memory)

- **RAM** is a temporary memory used while the computer is working.
- More RAM helps the computer open programs faster and run multiple tasks smoothly.

4. GPU (Graphics Processing Unit)

- The **GPU** creates the images, videos, and graphics you see on the screen.
- It is very important for gaming, video editing, and 3D applications.

5. HDD (Hard Disk Drive)

- The **HDD** stores your files, photos, videos, games, and operating system.
- It offers large storage space but is slower than an SSD.

6. SSD (Solid State Drive)

- The **SSD** also stores data, but it is much faster than an HDD.
- It makes the computer start quickly and loads games and applications faster.

7. PSU (Power Supply Unit)

- The **PSU** supplies electricity to all the computer components.
- Without it, the computer cannot turn on.

Q2. Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

**

Hint: Think about which one is faster, which is more shock-resistant, and where you would usually find each type (laptop, gaming PC, etc.). **

ANS.

Feature	HDD	SSD
1. Full Form	Hard Disk Drive	Solid State Drive
2. Storage Technology	Uses spinning magnetic disks	Uses flash memory chips (no moving parts)
3. Speed	Slower read/write speed	Much faster read/write speed
4. Durability	Less durable because it has moving parts	More durable because it has no moving parts
5. Shock Resistance	Easily damaged by shocks or drops	Highly shock-resistant
6. Power Consumption	Consumes more power	Consumes less power
7. Noise	Produces noise while operating	Silent operation
8. Cost	Cheaper and offers more storage for the price	More expensive but provides better performance
9. Typical Use Cases	Desktop PCs, external hard drives, backups, CCTV storage	Laptops, gaming PCs, operating systems, video editing
10. Best For	Users needing large storage at a low cost	Users needing fast performance, quick boot times, and smooth gaming

Q3. Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

ANS.

- **CPU (Processor):** This is the brain of your computer that handles the game's logic and rules. It tracks player movements, calculates bullet physics, and controls the computer-controlled enemies (bots)
- **RAM (Memory):** This acts as the temporary, super-fast workspace for your computer. It holds all the immediate game data, like the map and player's skin, so the system can open them instantly without lagging.
- **GPU (Graphics Card):** This is the artist that draws everything you see on your screen. It takes care of rendering the 3D graphics, buildings, explosions, and gunfire, so the game looks smooth and beautiful.

Q4. Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

ANS.

Hardware Component	Why Upgrade It?
CPU (Processor)	A faster CPU improves game performance and reduces lag by processing game data quickly.
GPU (Graphics Card)	A better GPU provides higher graphics quality, smoother gameplay, and better frame rates (FPS).
RAM	Increasing RAM (e.g., from 8 GB to 16 GB) allows the game and background applications to run smoothly.
SSD (Solid State Drive)	Installing the game on an SSD reduces boot time and speeds up game loading.
Motherboard	A newer motherboard supports the latest CPUs, RAM, and other upgraded components.
Power Supply (PSU)	A higher-wattage PSU provides enough power for upgraded hardware, especially a new GPU.
Cooling System	Better cooling prevents overheating and helps maintain stable performance during long gaming sessions.
Monitor	A monitor with a higher refresh rate (e.g., 144 Hz) makes gameplay smoother and more responsive.
Internet Adapter (Optional)	A faster Wi-Fi or Ethernet adapter improves online FIFA matches by reducing lag and latency.
Operating System & Drivers	Updating Windows and graphics drivers improves compatibility, stability, and overall game performance.